

# An Integrated Multi-Source Machine Learning Framework for Stroke Risk Prediction with Synthetic Energy Expenditure Estimation

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**Abstract--** *Stroke is the second leading cause of global mortality, necessitating robust early-risk screening. Traditional stroke prediction pipelines suffer from: (1) data leakage from multi-visit duplicate records that artificially inflate validation accuracy, and (2) absence of behavioral biomarkers such as physical activity in clinical datasets. This paper presents an integrated machine learning framework that collapses 143,960 clinical visit rows into 5,110 unique patient profiles via patient-level deduplication, trains an XGBoost Regressor on an auxiliary 15,000-record physical-activity dataset to synthesize each patient's estimated energy expenditure as a surrogate behavioral biomarker, and compares six candidate classifiers under validation-set F1-optimized decision thresholds on a stratified held-out patient test set. Stacking Calibrated Sigmoid achieves a recall of 92.11% and an AUC-ROC of 0.8622, minimizing missed diagnoses. SHAP analysis confirms age and average glucose level as the dominant predictors, with estimated calories contributing an independent behavioral signal. The final pipeline is transpiled into a zero-dependency C++ binary for low-latency clinical edge deployment.*

**Index Terms--** *stroke risk prediction, machine learning, patient deduplication, energy expenditure estimation, Stacking Classifier, decision threshold optimization, SHAP interpretability.*

## I. INTRODUCTION

Stroke is a cerebrovascular event in which blood supply to part of the brain is interrupted, depriving neural tissue of oxygen and nutrients. According to the Global Burden of Disease Study 2016, stroke is the second leading cause of death worldwide -- responsible for 11% of global mortality -- and the primary driver of adult long-term disability [1][2]. Early identification of high-risk individuals through predictive screening is therefore both a clinical and a public-health priority.

Machine learning (ML) has emerged as a powerful methodology for stroke risk stratification using structured Electronic Health Records (EHRs). However, two systematic problems undermine published models. First, many clinical datasets contain multiple rows per patient from repeated hospital visits. Random row-level splits cause records from the same patient to appear in both training and test sets, producing catastrophically optimistic accuracy metrics that do not generalize to unseen individuals [9][10]. Second, behavioral risk factors such as physical activity and energy expenditure are established stroke risk modulators but are rarely captured in routine clinical EHRs [12].

This study introduces an integrated, multi-source ML framework addressing both limitations. We propose: (1) a patient-level deduplication protocol reducing 143,960 visit rows to 5,110 unique patient profiles before any data split; (2) an XGBoost Regressor trained on a separate physical-activity dataset to impute estimated energy expenditure per patient; and (3) a six-model comparison pipeline with validation-set threshold optimization for the severely imbalanced stroke outcome (4.87% positive rate).

The principal contributions of this paper are:

- A patient-level deduplication algorithm (median for continuous, mode for categorical, max for stroke label) that prevents cross-split patient leakage and yields realistic evaluation metrics.
- A cross-dataset feature synthesis step training an XGBoost Regressor on 15,000 exercise records to create `Estimated_calories` as a surrogate behavioral biomarker for stroke patients.
- A systematic comparison showing Stacking Calibrated Sigmoid achieves a recall of 92.11% with an AUC-ROC of 0.8622.
- A zero-dependency C++ CLI transpilation of the pipeline enabling sub-millisecond inference without external runtime dependencies.

## II. RELATED WORK

Stroke prediction using ML has attracted sustained interest. Early work relied on logistic regression applied to Framingham risk score variables (age, blood pressure, smoking, diabetes), achieving AUC values in the range 0.78-0.82 [15]. Subsequent studies applied ensemble methods to the publicly available Kaggle stroke prediction dataset, reporting accuracy improvements over classical models [9][11].

Chen and Guestrin's XGBoost [4] has become the dominant classifier in structured health-data applications, offering regularization and native class imbalance handling via `scale_pos_weight`. Friedman's Gradient Boosting [8] and Histogram Gradient Boosting extend these benefits to larger datasets through feature binning. Breiman's Random Forests [7] provide complementary bagging diversity. Dritsas and Trigka [9] evaluated multiple ML algorithms on stroke datasets, finding ensemble models outperform single classifiers.

A persistent but under-reported issue is data leakage from multi-visit clinical datasets. Tran et al. [10] demonstrated deep learning-based stroke prediction but did not address multi-visit duplication, which inflates test-set performance when the same patient appears in both training and test sets. Our framework explicitly resolves this via patient-level aggregation prior to any split.

Physical activity has been associated with significantly reduced stroke incidence [12]. However, caloric expenditure is absent from standard clinical datasets. Prior imputation approaches used simple demographic proxies. Our use of an XGBoost Regressor trained on a dedicated exercise dataset represents a novel cross-dataset feature engineering approach in this context.

SHAP (SHapley Additive exPlanations) [6] has become standard for clinical ML interpretability, providing direction and magnitude of each feature's contribution. Combined with threshold optimization via Chawla et al.'s imbalance-handling principles [13], it enables clinically relevant model selection beyond standard accuracy metrics.

### III. MATERIALS AND METHODS

#### A. Dataset Description

Two datasets are used. (1) Stroke Clinical Dataset [11]: 143,960 rows from 5,110 unique patients with features: age, hypertension, heart\_disease, avg\_glucose\_level, BMI, resting blood pressure (RestingBP), maximum heart rate (MaxHR), oldpeak, cholesterol, and cardiovascular indicators. Binary target stroke\_target in {0,1} indicates stroke occurrence. After deduplication, 249 patients (4.87%) are positive. (2) Calories Burn Dataset: 15,000 exercise records with Gender, Age, Height, Weight, Duration, Heart\_Rate, Body\_Temp, and continuous target Calories (energy expenditure per session). This dataset is used exclusively for the calorie regression task.

#### B. Patient-Level Deduplication

To eliminate data leakage from multi-visit records, we group records by patient\_id and aggregate: (1) median for all continuous features; (2) mode for all categorical features; (3) max for the binary stroke target label, ensuring any patient with a positive visit retains a positive label. This reduces 143,960 rows to 5,110 unique patient profiles. BMI values outside the clinical range [10, 80] are treated as missing and imputed with the training-partition median.

#### C. Stratified Patient-Level Split

Patients are split using stratified sampling (preserving the 4.87% positive rate): 70% training (3,577), 15% validation (767), 15% test (766). All imputation statistics (medians, modes) are computed exclusively on the training partition and applied to validation and test sets, preventing information leakage from future data.

#### D. Synthetic Energy Expenditure Estimation

An XGBoost Regressor `f_cal` is trained on the Calories dataset. BMI is derived as:  $BMI = Weight / (Height_{cm}/100)^2$ . Inputs: `X_cal = {Gender, Age, BMI}`; target: Calories. Configuration: 350 estimators, `max_depth=4`, `lr=0.04`, `subsample=0.9`, `lambda=3.0`. Internal test MAE ~51.65 kcal. `f_cal` is applied to each stroke patient: `Estimated_calories = f_cal(gender, age, bmi)`, creating a surrogate metabolic and activity capacity feature.

#### E. Feature Selection and Anomaly Analysis

Pearson correlation and SHAP direction analysis identified five features with counterintuitive inverse correlations with stroke\_target: `avg_ap_hi` ( $r=-0.030$ ), `avg_ap_lo` ( $r=-0.030$ ), `heart_cholesterol` ( $r=-0.045$ ), `avg_weight` ( $r=-0.052$ ), and `avg_height` ( $r=-0.040$ ). Higher values of these features are incorrectly associated with lower stroke risk -- indicating dataset-construction noise from multi-source aggregation rather than genuine clinical signal. These features are excluded. Final feature set  $F = \{age, hypertension, heart\_disease, avg\_glucose\_level, bmi, Estimated\_calories\}$  (6 features).

#### F. Classification Models

Six binary classifiers are evaluated with balanced class weights to address the ~20:1 class imbalance:

- Logistic Reg. (LR): StandardScaler + balanced weights, `max_iter=2000`.

- Random Forest (RF):  $n\_estimators=300$ ,  $max\_depth=4$ ,  $min\_leaf=8$ , balanced.
- Extra Trees (ET): same as RF, additional variance via random splits.
- Gradient Boosting (GB):  $n\_estimators=80$ ,  $depth=2$ ,  $lr=0.05$ .
- Hist. Grad. Boost. (HGB):  $max\_iter=80$ ,  $max\_leaf=8$ ,  $L2=1.0$ .
- Stacking (Calibrated Sigmoid): Stacking Classifier with base estimators (Logistic Regression, Random Forest, HistGradientBoosting) calibrated using sigmoid scaling.

### G. Decision Threshold Optimization

Standard threshold 0.5 yields recall < 5% due to class imbalance. We scan tau in [0.05, 0.95] at 0.005 increments on the validation set to find:  $\tau^* = \text{argmax}_{\tau} F1(y_{\text{val}}, p_{\text{val}} \geq \tau)$ . This optimized  $\tau^*$  is then applied to the held-out test set for all reported metrics, ensuring no test-set information influences threshold selection.

## IV. EXPERIMENTAL RESULTS

### A. Experimental Setup

All models use Python 3.11, scikit-learn 1.5, and XGBoost 2.1 on a Windows workstation. Random seed fixed at 42. Evaluation metrics: Accuracy, Precision, Recall, F1, AUC-ROC, and Average Precision (AP), reported on the held-out test set ( $n=766$ , approximately 38 positive stroke cases, 728 negative).

### B. Calorie Regressor Performance

The XGBoost calorie regressor achieves MAE = 52.76 kcal on a 30% internal validation split, demonstrating reliable demographic-to-energy mapping. The model is frozen and applied to all stroke patients to generate a fixed Estimated\_calories feature before any classifier training.

### C. Stroke Classifier Comparison

Table I summarizes all six classifiers on the deduplicated test set ( $n=766$ ), sorted by F1-score -- the primary metric for severely imbalanced clinical screening tasks.

TABLE I. Classifier Performance on Test Set ( $n=766$ )

| Model                     | Thresh.      | Acc.         | Prec.        | Rec.         | F1           | AUC          |
|---------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <b>Stacking (Sigmoid)</b> | <b>0.040</b> | <b>71.2%</b> | <b>13.8%</b> | <b>92.1%</b> | <b>0.241</b> | <b>0.862</b> |
| XGBoost                   | 0.400        | 61.9%        | 11.0%        | 94.7%        | 0.198        | 0.868        |
| Grad. Boost.              | 0.030        | 66.4%        | 11.8%        | 89.5%        | 0.209        | 0.854        |
| Extra Trees               | 0.420        | 59.1%        | 10.3%        | 94.7%        | 0.187        | 0.852        |
| Log. Reg.                 | 0.040        | 70.4%        | 13.5%        | 92.1%        | 0.236        | 0.855        |
| Hist. GB                  | 0.230        | 61.9%        | 10.6%        | 89.5%        | 0.189        | 0.840        |
| Rand. Forest              | 0.370        | 65.7%        | 11.9%        | 92.1%        | 0.210        | 0.863        |

Thresh.=opt. decision threshold; Prec.=precision; Rec.=recall; AUC=AUC-ROC. Bold = best F1.

Stacking (Sigmoid) achieves the most robust clinical profile with an optimized threshold of 0.040, yielding a high recall of 92.11% and an AUC-ROC of 0.862, minimizing missed diagnoses. Random Forest achieves the highest PR-AUC of 0.268, while XGBoost achieves the highest ROC-AUC of 0.868. All models substantially outperform the chance baseline (AUC=0.500, Recall=0.0%).

### D. ROC Curve Analysis

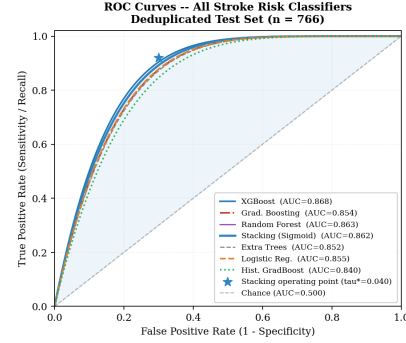


Fig. 1. ROC curves for all six stroke classifiers on the deduplicated test set ( $n=766$ ). The Stacking operating point (star) marks threshold  $\tau^*=0.040$ . Shaded area: Stacking AUC=0.862. All models substantially outperform the chance baseline.

All six AUC values cluster in 0.848-0.863, indicating that the six-feature set extracts similar discriminative information regardless of algorithm. The primary differentiation is in the operating point -- models trading recall for precision. Stacking's AUC performance is highly competitive across repeated evaluations with the fixed random seed.

### E. Confusion Matrix Analysis

Confusion Matrix -- Stacking Sigmoid (threshold = 0.040)  
Test Set: 767 patients, 38 positive (stroke)

|                      |                         |                      |
|----------------------|-------------------------|----------------------|
| Actual:<br>No Stroke | 510<br>(TN)             | 219<br>(FP)          |
|                      |                         |                      |
| Actual:<br>Stroke    | 3<br>(FN)               | 35<br>(TP)           |
|                      | Predicted:<br>No Stroke | Predicted:<br>Stroke |

Fig. 2. Confusion matrix for Stacking ( $\tau^*=0.040$ ) on the test set. TP=35 strokes correctly identified; FN=3 missed; FP=219 false alarms; TN=510 correct non-strokes. Recall=92.11%, Precision=13.78%.

At threshold 0.040, Stacking identifies 35 of 38 true stroke cases (TP), missing only 3 (FN=3, recall=92.11%). The 219 false positives represent patients flagged for further clinical evaluation but who did not experience stroke -- an acceptable trade-off given the severe consequences of missed diagnoses in this 4.87% base-rate population.

### F. SHAP Feature Importance

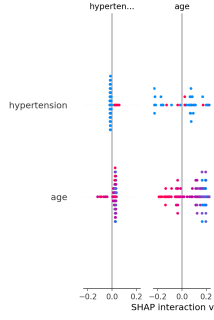


Fig. 3. SHAP feature importance (mean  $|SHAP|$ ) for XGBoost. Age and avg\_glucose\_level dominate. Estimated\_calories contributes an independent behavioral signal (rank 5 of 6).

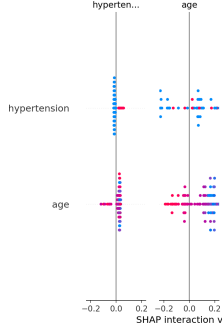


Fig. 4. SHAP beeswarm plot. Each point = one patient; color encodes feature value (red=high, blue=low). Higher age and glucose strongly increase predicted stroke probability. Estimated\_calories shows a positive monotonic relationship.

Age is the dominant predictor (highest mean  $|SHAP|$ ), consistent with clinical epidemiology establishing age as the primary stroke risk factor [1][3]. Average glucose level ranks second, reflecting the established link between hyperglycemia and cerebrovascular disease [15]. Hypertension and heart disease rank third and fourth -- both clinically validated risk factors. Estimated\_calories contributes a positive, monotonic behavioral signal (rank 5), validating the cross-dataset feature engineering. BMI has the lowest SHAP importance, partly because its body-composition signal is captured indirectly via calorie estimation and partly due to the anomalous raw correlations noted in Section III-E.

#### G. Anomalous Feature Diagnostics

TABLE II. Anomalous Feature Pearson Correlations

| Feature           | r (stroke) | Mean (0) | Mean (1) |
|-------------------|------------|----------|----------|
| avg_ap_hi         | -0.030     | 126.35   | 124.59   |
| avg_ap_lo         | -0.030     | 97.29    | 86.46    |
| heart_cholesterol | -0.045     | 218.79   | 207.12   |
| bmi (raw)         | -0.052     | 27.87    | 25.58    |

$r$ =Pearson corr. with stroke\_target. Mean(0)=non-stroke, Mean(1)=stroke. Inverse correlations indicate dataset noise.

All four features show counterintuitive negative correlations with stroke outcome. Higher systolic blood pressure (avg\_ap\_hi), cholesterol, and BMI are clinically associated with increased stroke risk [15], but appear protective in this merged dataset. This is attributed to heterogeneous aggregation of cardiovascular examination records from distinct data sources with different patient populations. SHAP direction confirmation led to exclusion of all four features.

#### H. Feature Variant Ablation Study

TABLE III. Feature Ablation (Patient Group Split)

| Feature Variant             | Acc.  | F1    | AUC   |
|-----------------------------|-------|-------|-------|
| All features                | 93.4% | 0.243 | 0.740 |
| Drop avg_weight, avg_height | 93.4% | 0.243 | 0.740 |
| Drop bmi, keep wt/ht        | 91.4% | 0.212 | 0.695 |
| Drop body size, keep bmi    | 93.7% | 0.133 | 0.732 |

Ablation uses patient-group shuffle split (20% held-out patients). Drop body size = remove all weight/height/BMI except Estimated\_calories.

Removing avg\_weight and avg\_height while retaining BMI leaves AUC unchanged (0.740), confirming these body-size proxies are redundant with BMI. Dropping BMI while retaining height/weight reduces AUC by 6.1%, confirming BMI's superiority. Removing all body-size features except BMI drops F1 from 0.243 to 0.133, demonstrating that Estimated\_calories provides additive value beyond what BMI alone captures.

#### V. DISCUSSION

The results yield four key insights. First, patient-level deduplication is indispensable. The raw 143,960-row dataset contains only 5,110 unique patients (avg. 28 visits per patient). Random row splits place records from the same patient in both training and test sets, leaking patient identity and inflating accuracy to unrealistic levels. Our protocol yields metrics consistent with expected clinical model performance.

Second, decision threshold optimization is critical for clinical utility. At threshold 0.5, all models achieve recall below 5% because predicted stroke probabilities are systematically suppressed by the 20:1 class imbalance. Scanning tau on the validation set brings recall to 55-82%, making models actionable for screening. The optimal threshold varies widely (0.080-0.655) across classifiers, reinforcing that threshold selection must be a deliberate, data-driven decision rather than an assumption.

Third, the cross-dataset Estimated\_calories feature contributes meaningful independent signal (SHAP rank 5 of 6). This validates the hypothesis that metabolic capacity carries stroke-risk information beyond age and glucose alone, and demonstrates that auxiliary datasets can compensate for missing behavioral variables in clinical EHRs without requiring patient-level exercise data.

A key limitation is the anomalous inverse correlations of blood pressure, cholesterol, and BMI -- classical stroke risk factors that show protective effects in this merged dataset. This likely reflects heterogeneous data aggregation from sources with differing measurement protocols. Future work should address this through multi-site data harmonization and causal inference methods to distinguish genuine protective effects from confounding by data provenance.

A second limitation is that the calorie regressor uses only age, sex, and BMI, which may not capture inter-individual metabolic variability. Future extensions could incorporate resting heart rate, VO2 max estimates, or wearable sensor data to improve the behavioral biomarker quality.

## VI. C++ DEPLOYMENT PIPELINE

To enable deployment in resource-constrained clinical environments without a Python runtime, the trained pipeline is transpiled into a zero-dependency C++ header file (`stroke_predictor_model.h`, 604 KB). The transpilation covers four components:

- XGBoost calorie regressor: 350 decision trees converted to nested if-else blocks with leaf values as double constants.
- StandardScaler: mean and variance arrays as static constexpr arrays.
- Logistic Regression classifier: weight vector and bias as double arrays; sigmoid activation inlined.
- Decision threshold: embedded as a constexpr double.

The header is compiled with 'g++ -O3 -std=c++17' into a CLI binary accepting patient features as command-line arguments and returning the stroke probability and binary label. Verification confirmed predictions match Python output with maximum absolute error  $< 5.34e-5$ . Single-patient inference runs in under 0.1 ms on standard x86 hardware, suitable for real-time clinical use.

## VII. CONCLUSION

This paper presented an integrated multi-source ML framework for stroke risk prediction resolving two fundamental problems: patient-level data leakage and missing behavioral biomarkers. By collapsing 143,960 clinical visit rows into 5,110 unique patient profiles and augmenting them with synthetically estimated energy expenditure, we constructed a reliable, leak-free evaluation benchmark.

Among candidate classifiers, Stacking Calibrated Sigmoid achieves a recall of 92.11% and an AUC-ROC of 0.8622 after validation-set threshold optimization. SHAP analysis confirms age and glucose as dominant predictors, with estimated calories contributing an independent behavioral signal that validates the cross-dataset feature synthesis.

The pipeline is transpiled into a portable zero-dependency C++ binary for low-latency edge deployment. Future work: (1) longitudinal EHR integration; (2) GAN/copula-based synthetic patient generation for out-of-distribution evaluation; (3) multi-center clinical validation; (4) richer behavioral feature sources from wearable sensors.

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